

Clinical Readmission Analytics & Risk Prediction Model Report

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DSC-26 Machine Learning

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Project Documentation & Technical Report

Overall Summary:

In corporate hospitals who provide services on insurances are facing high losses due to readmissions of diabetes patients within less than 30 days of discharge from hospital. My model will help the hospital management to reduce their losses by estimating which patients have high chances of readmission within 30 days and following up with patient health condition on time with help of a coordinator.

Dataset:

I have developed this machine learning model using **Diabetes 130-US Hospitals Dataset** (covering 10 years of clinical care across 130 US hospitals). This dataset has 101,766 records with 50 features encompassing demographics, admission attributes, clinical metrics, diagnoses, lab results, and 23 diabetes medications.

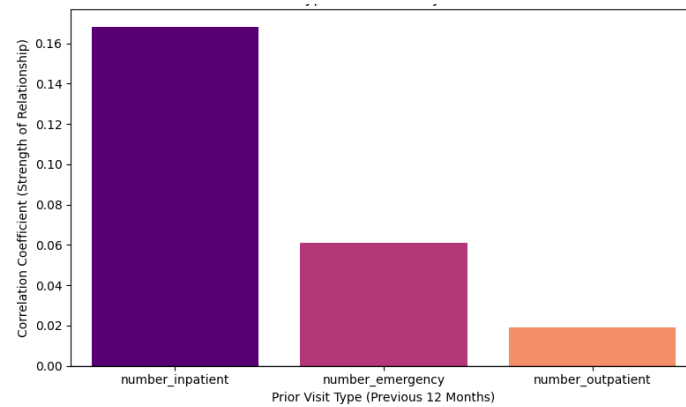
Handling missing values:

- Dropped feature 'weight' because of 96% null values and this feature doesn't impact much while analysing patient chances of readmission.
- For 'payer_code' and 'medical_speciality' replaced null with 'unknown'.
- For features like 'max_glu_serum' and 'A1Cresult' imputed missing values as 'Not Measured'
- Didn't remove any row from the data set because this is a patient related data set and removing any row implies removing patient whole record so I performed only column level cleaning to avoid any loss of data.

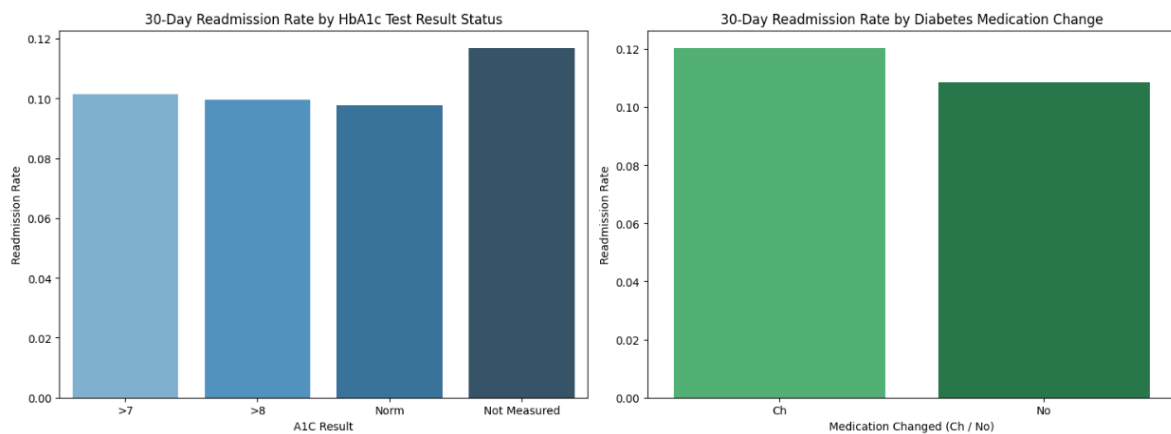
Exploratory Data Analysis (EDA) summary:

Target Variable Distribution (30-day Readmission):
Non-Readmitted (0): 88.61%
Readmitted <30 Days (1): 11.39%

- Early readmission with 11.39% of total encounter clearly shows imbalanced classification problem here.



- In patients count showed strongest correlation with readmission when compared to emergency patients and out patients in the last twelve months.



- Patients with >7 elevated HbA1c results has more readmission rate even most of the data has no data for A1c results from available results I could see this.
- Patients who medication has been changed shows high readmission rate.
- Removed features 'exmaide and 'citoglipton' to reduce unwanted dimensionality as they have constant values across the data.

Feature engineering and Pre processing:

- Transformed age intervals (eg.,[70-80]) to continuous numerical midpoints([75]) and I dropped 'age' feature because keeping both columns will create confusion and leads mis interpretation by the model.
- Created a new measure 'total_health_system_visits' by summing up 'number_outpatient' + 'number_emergency'+ 'number_inpatient'.
- Created a binary flag to check if total number of prescribed medication exceeds 10.
- Re-encoded ICD-9 diagnosis codes of feature 'diag1' into 9 categories like Circulatory, Digestive, Respiratory, Diabetics, Injury, Neoplasms, Musculoskeletal, Genitourinary and other.
- Partioned dataset using 80/20 train-test split at patient level to ensure multiple visits of a patient never leaks across train and test .

- Using Scikit-learn for numerical columns I imputed missing values with median and then applied scaling ; for categorical columns imputed missing values with 'Missing' and then applied one-hot encoding.

Modelling and Hyperparameter tuning:

I have trained cleaned dataset using three different models and used manual or randomizedsearchcv tuning techniques based on the model to identify which model gives better output.

Model 1: Logistic Regression (Baseline)

-Reason: I choosed this as it is highly interpretable, Computationally light weight, naturally generates fully calibrated risk scores between 0 and 1 and no further scaling is required.

-Tunning strategy: I picked this as it is low search space dimentionality and testing a handful values along class weight balanced manually allows us to quickly identify the optimal regularization penalty without the overhead of running automated search frameworks.

Model 2: Random Forest Classifier (Bagging Ensemble)

-Reason:

I choosed this because its resistance to overfitting, it requires minimal tunning, helps to handel noisy data well and simplifies structural tracking by claculating exactly which variable reduces data impurity the most.

-Tunning strategy: I picked this as it has high-dimensional parameter grid and RandomizedSearchCV samples a fixed no.of parameters setting from specified distribution. This statistically uncovers near- optimal parameters combinations in a fraction of the computational time.

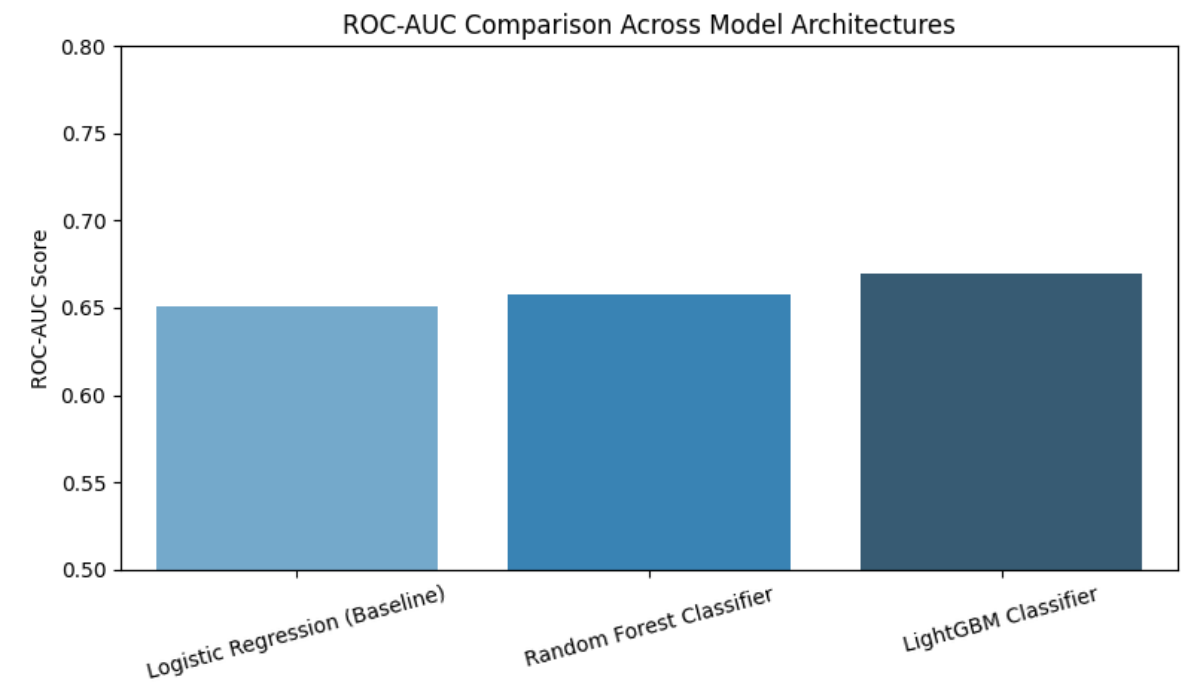
Model 3: LightGBM Classifier (Gradient Boosting)

-Reason: Requires minimal RAM allocation even when processing millions of rows of data. My data set has more than a million of data rows. So I thought I can try this.

-Tunning: I picked manual testing for this as it provides specific imbalance mechanism and controlled tree complexity.

Performance Evaluation and Model Comparison:

To analyse which model gives best performance I have evaluate following metrics ROU-AUC, Precision, Recall and F1 Score.



LightGBM classifier gave best performance with following score to each metrics:

ROC-AUC:0.669841

Precision:0.177668

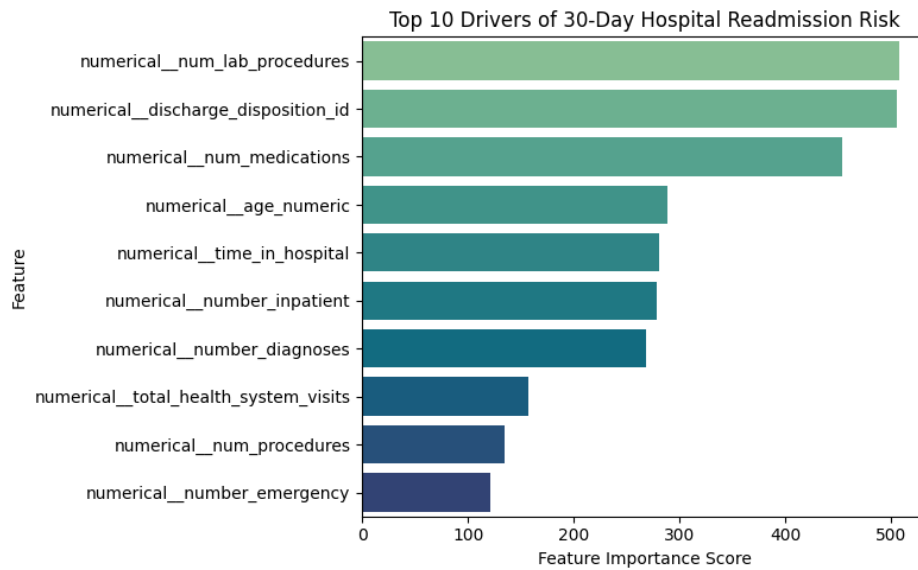
Recall:0.576014

F1 Score:0.271571

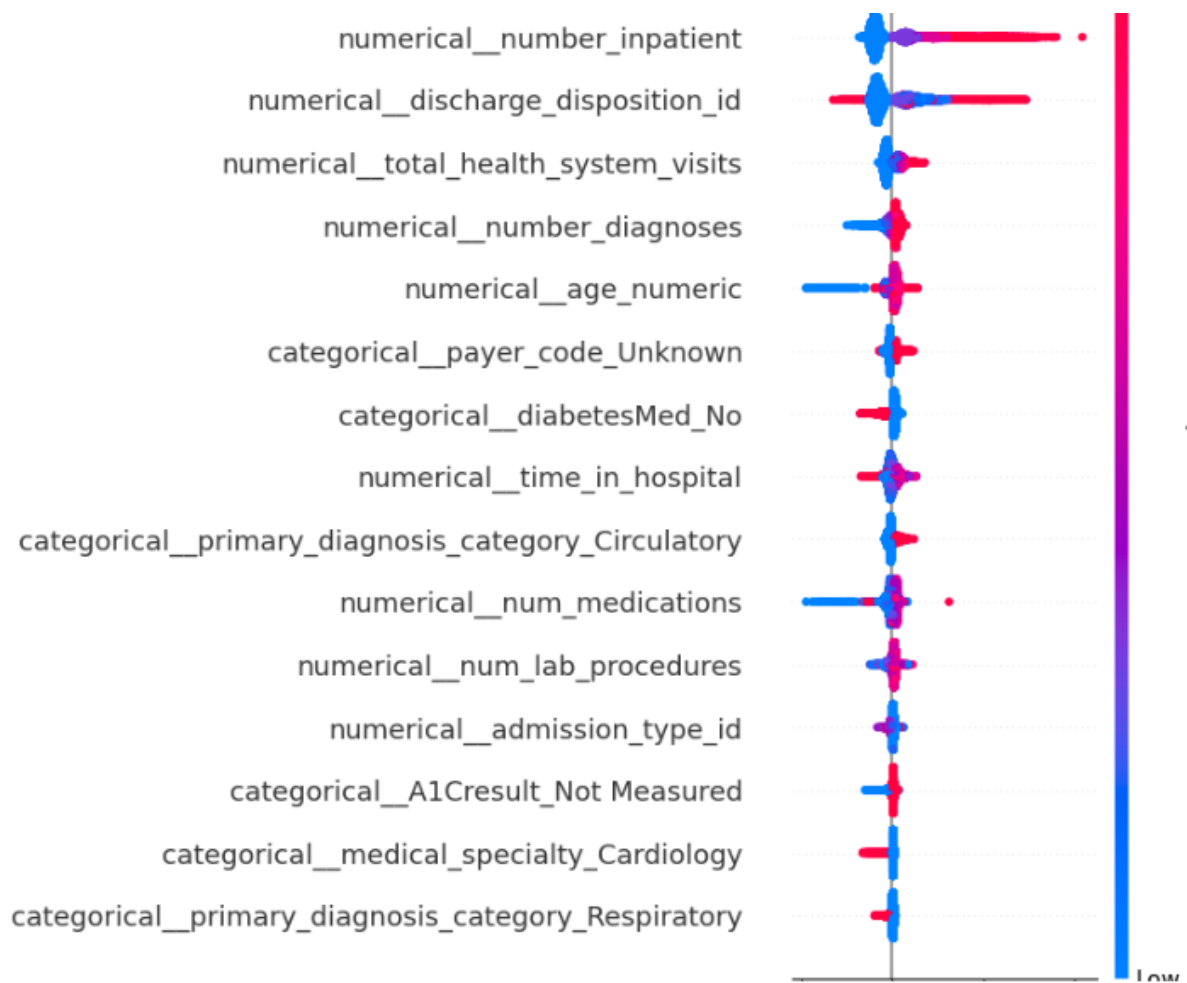
While other scores are like this;

	Model Architecture	Tuning Strategy	Test ROC-AUC Score	Test Recall (Class 1)	Test Precision (Class 1)	Test F1-Score (Class 1)
0	Logistic Regression (Baseline)	Manual Hyperparameter Tuning (C=0.1, class_wel...	0.650959	0.524744	0.175803	0.263370
1	Random Forest Classifier	RandomizedSearchCV (10 Iterations, 3-Fold CV)	0.657302	0.523406	0.176781	0.264295
2	LightGBM Classifier	Manual Hyperparameter Tuning (learning_rate, s...	0.669841	0.576014	0.177668	0.271571

Feature Importance and SHAP:

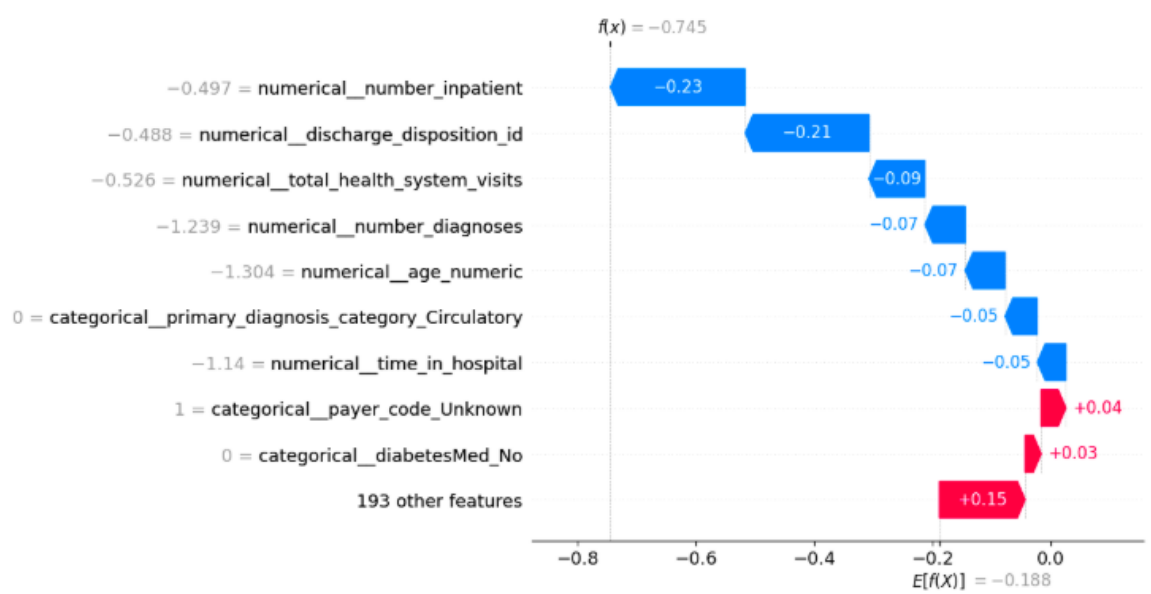


From the result numerical_num_lab_procedures and numerical_discharge_disposition_id are two overall high performance features so that we can say this model is highly relies on diagnostics and historical hospital usage to measure the risk.



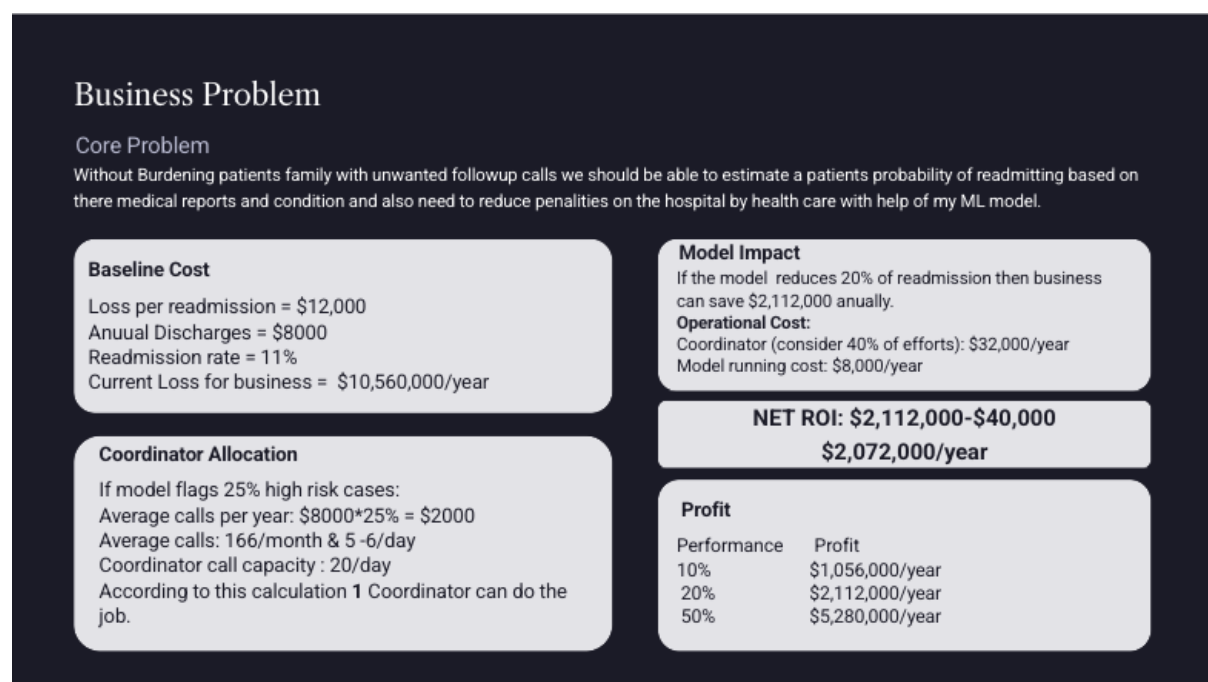
From the above result I understood that prior hospital stays and complex discharges are the biggest red flags for readmission.

This



This graph helps us to understand how model the classify a patient into different level. Based on the model if has history for low hospital visits and low overall illness complexity.

HOW MY ROI MODEL LINKS TO THIS RESULTS and MANAGEMENT DECISION:v



False positive: This model predicts a patient will be readmitted but they will not return.

False Negative: This model predicts a patient will not be readmitted but they may end up back in the hospital.

According to my ROI model problem is - Without Burdening patients family with unwanted follow up calls we should be able to estimate a patients probability of readmitting based on their medical history and health condition and also need to reduce penalties on the hospital by health care authorities with help of my ML model.

By keeping this in mind my whole notebook focuses on prediction threshold rather than basic 0/1 outputs. So my model predicts a precise probability score that helps to segregate patients into high level to low level risk.

I didn't used default 0.50 probability threshold in code. Instead I tuned threshold to 25% highest risk predicted patients. This bridges gap between Machine Learning and Operational Costs.

If my model captures 60-70% + of actual readmission with in top 25% cohort, a 20% reduction via intervention is fully achievable. So our estimated profit can be achieved.

If a false positive scenario happens loss will be approximately \$15 to \$20 and few mins of time of coordinator which is very less.

If a false negative scenario happens loss will be \$12,000 in form of penalties to health insurance companies which is huge.

I will say this model is good to deploy.

SELF REFLECTION:

During this project I have learned some new things:

- 1.What is complete life cycle of a machine learning model.
- 2.Importance of understanding a business before starting the code.
- 3.Even using AI is not that is the whole trick lies in prompting I have got opportunity to practice that in this project. I don't say that I haven't used in other projects but doing a project alone and figuring out everything on my own what to ask, when to ask like after which step about which step I should ask about I got clear.
- 4.For every step in lifecycle I have got new challenges while decoding those and finding answers I understood the link between each step.
- 5.Also learned in how many types I can load a dataset from any website.

Major challenge I faced during this project completely changed my approach towards the project execution and this is also one of the reason for not trying more than 3 models. Initially without life cycle I thought understanding business problem could alone help to build the model but my knowledge about ML life cycle is very poor and in last day of making report I has to change everything, actually I have built model without analysing feature importance and SHAP and I thought project is completed after training 3 models and figuring out which best but then when I am cross checking notes given you and online I got understand importance of these two

steps and also connecting my ROI estimate and notebook results step also I felt hard but finding connect made me understand how to work for a real business . So, from this I learned basics are more important than process. Now I am confident about the process may be technically I might made mistakes, but I got confidence that I can handle any more model building alone and my research and understanding skills are also improved a lot.

It's a great experience and best push to my early career thankyou professor. Your guidance and notes helped alot while doing project along with AI.